

Team Division — Campus Library Book Registry

Project Idea 6 · Technologies of Cryptocurrencies · 5 Team Members

Member 1

Blockchain Engineer

Core Smart Contract

structs, mappings, require validation

Access Control Module

onlyOwner modifier, getAdmin() view function

Batch Admin Operations

array parameters, for-loop, require checks

Pause & Resume Switch

paused flag, whenNotPaused modifier

Ownership Transfer (Solidity)

transferOwnership function

Member 2

Coin & Deployment

Custom ERC-20 Coin

Library Coin contract, mint restricted to admin

Auto-Setup Script

deploy both contracts, seed 3+ fake books

User Registration (Solidity)

mapping(address => string), registerUser function

Balance Snapshot CSV

scan blocks, write CSV with balances

System Documentation

README.md — setup, admin guide, user guide

Member 3

Terminal App

Terminal App (main menu)

user menu + hidden admin gate by password/key

Transaction Sender

send_transaction, receipts, error handling

User Registration (UI side)

first-visit prompt, display name in header

Coin & ETH Balance Checker

balanceOf + get_balance, side-by-side display

Member 4

Data & Analytics

Admin Dashboard Script

block scan, coins minted, top-3 active users

Data History Report

most borrowed books table from chain history

Personal Activity History

scan blocks by address, print action table

Ownership Transfer (Python)

transfer + re-test script showing access changes

Member 5

Security & Monitoring

Security Testing

onlyOwner revert test, automated PASS/FAIL output

Live Alert System

polling loop, detect borrow/return events

Integration Testing

end-to-end smoke tests across all features

Final Setup & Demo Prep

Ganache config, run guide, demo walkthrough

Core rules (all members): Every project enforces Admin/User role separation. All write operations use the **onlyOwner** modifier. The custom ERC-20 Library Coin can only be minted by the Admin. A terminal app is required; a GUI is an optional bonus.